Rates of Change & Concavity

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.

2 [Foundational] • Target: Average rate of change

A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.

3 [Foundational] • Target: Increasing/decreasing behavior from rate of change

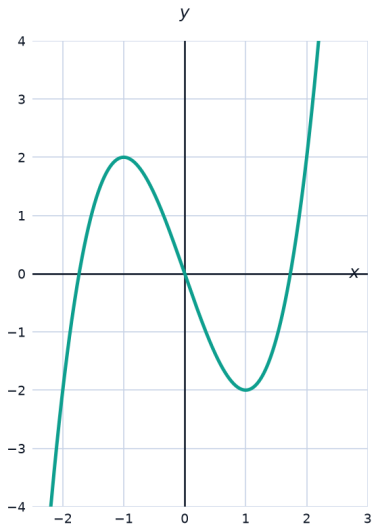
A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.

4 [Foundational] • Target: Increasing/decreasing behavior from rate of change

A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.

5 [Foundational] • Target: Visualizing concavity and inflection

The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.



a Describe the concavity of the graph on $(-2, 0)$.

b Describe the concavity of the graph on $(0, 2)$.

c State what kind of point $x = 0$ represents.

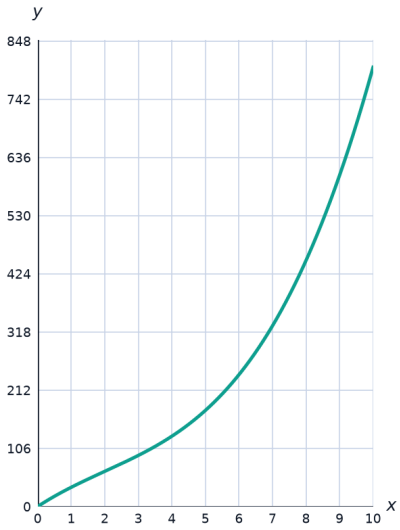
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6 [Foundational] • Target: Rate of change of the rate of change

Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

7 [Foundational] • Target: Applying rate-of-change reasoning to models

A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \le t \le 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applying rate-of-change reasoning to models

For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

9 [Intermediate] • Target: Estimating rate of change from a table

A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

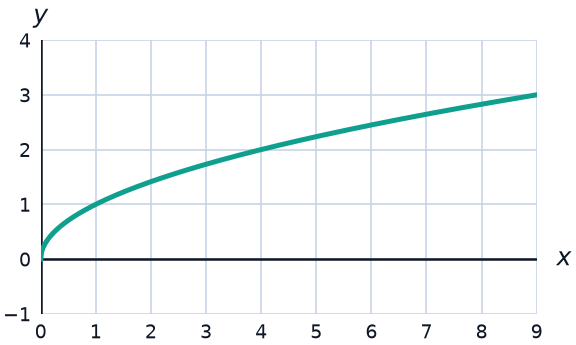
10 [Intermediate] • Target: Estimating rate of change from a table

Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

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11 **[Intermediate]** • **Target: Visualizing rate of change on a graph**

The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.



a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$.

b Explain how the shrinking steepness in the graph relates to your two AROC values.

c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern.

12 **[Intermediate]** • **Target: Concavity and real-world interpretation**

A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

13 **[Intermediate]** • **Target: Applying rate-of-change reasoning to models**

Find the average rate of change of the polar radius $r(\theta) = 3 + 2\sin(2\theta)$ over the interval $[0, \frac{\pi}{4}]$, and interpret whether the distance from the origin is increasing or decreasing on this interval.

14 **[Intermediate]** • **Target: Estimating rate of change from a table**

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in x?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Visualizing rate of change on a graph** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta + 3\sin\theta - 3 = 0$ on the interval $[0, 2\pi)$.

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16 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 4.03 Parametric Equations & Modeling

Level 2 · MediumSolve the logarithmic equation $\log_2(x+3) + \log_2(x-1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.

17 [Challenging] • Target: Average rate of change • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A bivariate dataset (x, y) is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y}) = 1.5 + 0.4x$. Express $\hat{y}$ as an exponential function of x in the form $\hat{y} = a \cdot b^x$, and find the predicted percentage growth rate per unit increase in x.

18 [Challenging] • Target: Increasing/decreasing behavior from rate of change • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where t is measured in days. Determine the exact daily multiplicative decay factor b in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

19 [Challenging] • Target: Visualizing concavity and inflection • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Solve the exponential equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$ for all real x.

20 [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Level 1 · FoundationsLet $q(x) = -(x+2)^2(x-1)(x-4)^3$. What is the degree of $q(x)$, does the graph cross or touch the x-axis at $x = -2$, and what is $\lim_{x \to \infty} q(x)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.

a. 35

b. -21

c. 33

d. 37

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22 [Foundational] • Target: Increasing/decreasing behavior from rate of change

A car's odometer reading (miles) is recorded every hour: $t = 0$: 131; $t = 1$: 150; $t = 2$: 178; $t = 3$: 215. Is the car's speed increasing, decreasing, or constant?

a. Decreasing

b. Constant

c. Cannot be determined from this data

d. Increasing

23 [Foundational] • Target: Visualizing concavity and inflection

A function h has average rates of change of 4 on $[0, 1]$, 2 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

a. Concave up

b. Cannot be determined from this data

c. Linear (neither concave up nor down)

d. Concave down

24 [Foundational] • Target: Rate of change of the rate of change

The rate of change of a quantity N is proportional to its current value, modeled by the differential equation $\frac{dN}{dt} = kN$, where k is a constant. If the initial population at time $t = 0$ is N_0, and at time $t = 5$ years, the population has grown to $2N_0$, what is the population at time $t = 15$ years, expressed in terms of N_0?

a. $6N_0$

b. $8N_0$

c. $7N_0$

d. $10N_0$

25 [Foundational] • Target: Applying rate-of-change reasoning to models

If $2 + 3i$ is a root of a real polynomial, so is:

a. $2 - 3i$

b. $-2 + 3i$

c. $-2 - 3i$

d. $3 + 2i$

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26 [Foundational] • Target: Estimating rate of change from a table

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the x-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose x-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of k?

a. 400

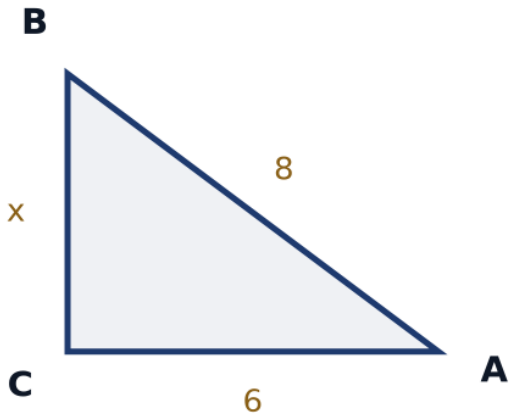
b. 225

c. 256

d. 625

27 [Foundational] • Target: Visualizing rate of change on a graph

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?



a. $\frac{3}{5}$

b. $\frac{4}{5}$

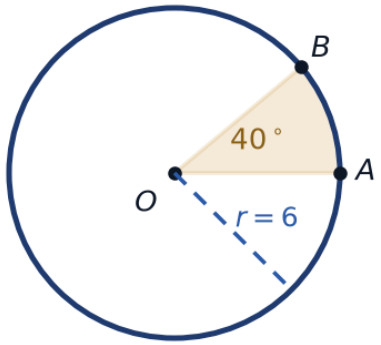
c. $\frac{3}{4}$

d. $\frac{4}{3}$

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28 [Foundational] • Target: Concavity and real-world interpretation

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{1}{2}$

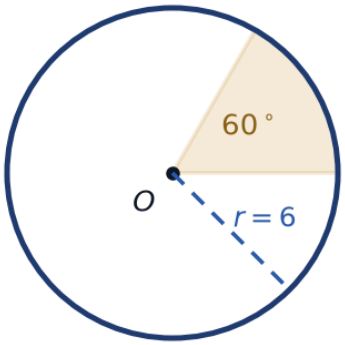
d. $\sqrt{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Average rate of change**

The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates (x, y) of point P?



a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

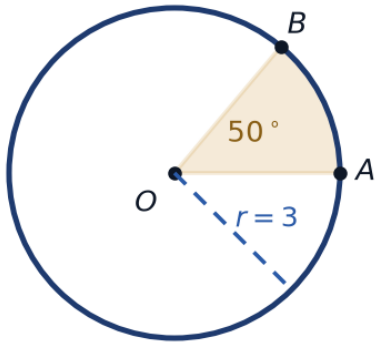
c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

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30 **[Intermediate]** • **Target: Increasing/decreasing behavior from rate of change**

A point P is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point P?



a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

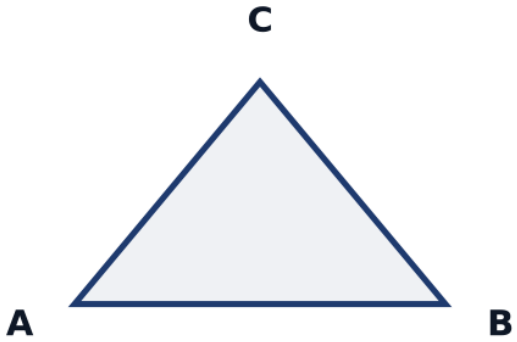
b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

31 **[Intermediate]** • **Target: Visualizing concavity and inflection**

$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:



a. $\frac{3}{4}$

b. $\frac{4}{3}$

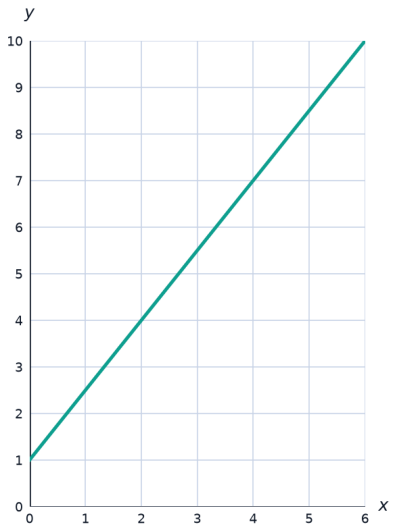
c. $\frac{3}{5}$

d. $\frac{5}{3}$

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32 [Intermediate] • Target: Rate of change of the rate of change

In polar form, which equation represents a circle of radius 3 centred at the origin?



a. $r = 3$

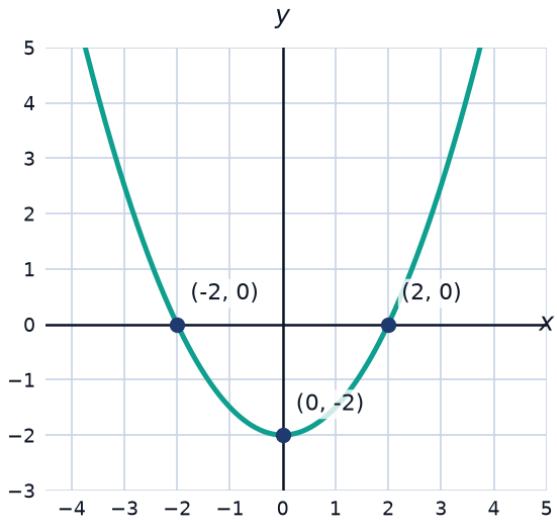
b. $r = 3\theta$

c. $r = 3\cos\theta$

d. $\theta = 3$

33 [Intermediate] • Target: Applying rate-of-change reasoning to models

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?



a. 3

b. 4

c. 6

d. 8

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34 [Intermediate] • Target: Estimating rate of change from a table

A particle's position in the Cartesian plane is described by the polar coordinates (r, θ), where r is the distance from the origin and θ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$?

a. 13

b. 25

c. 49

d. 52

35 [Intermediate] • Target: Visualizing rate of change on a graph

Consider a particle whose position in the complex plane at time t (in seconds) is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. (a) Find the initial position of the particle at $t = 0$ in polar coordinates (r, θ), where $r \geq 0$ and $0 \leq \theta < 2\pi$. (b) Determine the angular velocity of the particle in radians per second. (c) If the particle starts at a point A and moves along a circular path, find the arc length traveled by the particle from $t = 0$ to $t = 4$ seconds.

a. (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) 2π

b. (a) $(3, \frac{\pi}{6})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{4\pi}{3}$

c. (a) $(3, \frac{2\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) $\frac{2\pi}{3}$

d. (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{8\pi}{3}$

36 [Intermediate] • Target: Concavity and real-world interpretation

A vector $\mathbf{v}$ is defined by its component form $\langle 2a + 1, 3 - b \rangle$. This vector is then transformed by a translation represented by the vector $\mathbf{t} = \langle 5, -2 \rangle$. The resulting transformed vector, $\mathbf{v}'$, is given as $\langle 12, 1 \rangle$. Determine the values of the parameters a and b.

a. a = 3, b = -4

b. a = 4, b = 3

c. a = 3, b = 4

d. a = 4, b = -3

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37 [Intermediate] • Target: Average rate of change

A point P has polar coordinates (r, θ). When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle θ is restricted to the interval $[0, 2\pi)$, find the value of r and θ in radians.

a. $r = 6, \theta = \frac{\pi}{3}$

b. $r = 3\sqrt{2}, \theta = \frac{\pi}{4}$

c. $r = 6, \theta = \frac{\pi}{6}$

d. $r = 3, \theta = \frac{\pi}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Increasing/decreasing behavior from rate of change • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Consider a linear transformation T in the plane represented by the matrix $M = (a, b; c, d)$. A triangle with vertices at $A = (1, 2)$, $B = (3, 1)$, and $C = (2, 4)$ is transformed by T to a new triangle $A'B'C'$. If the area of the original triangle ABC is 3 square units and the area of the transformed triangle $A'B'C'$ is 18 square units, and given that $a = 2$ and $d = 3$, find the product of the possible values of the determinant of M.

a. 6

b. -6

c. 36

d. -36

39 [Challenging] • Target: Visualizing concavity and inflection • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

a. $\frac{100}{26}$

b. $\frac{240}{26}$

c. $\frac{576}{26}$

d. $\frac{600}{26}$

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40 [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is θ. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius R, find the value of R.

a. $\frac{5}{2}$

b. 5

c. $\frac{5\sqrt{3}}{2}$

d. 10

41 [Challenging] • Target: Applying rate-of-change reasoning to models • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

A transformation is applied to a triangle ABC, mapping it to triangle A'B'C'. The transformation can be represented by a matrix multiplication of the form $M(x; y) = (3, 1; 1, k)(x; y)$, where M is the transformation matrix and k is an unknown parameter. If the original point A is $(2, 1)$ and its transformed image A' is $(7, 5)$, find the value of k.

a. 2

b. 3

c. 4

d. 5

42 [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

The terminal ray of an angle θ in standard position in the xy-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$?

a. $-\frac{5}{13}$

b. $\frac{12}{13}$

c. $-\frac{12}{5}$

d. $\frac{12}{5}$

43 [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 4.01 Inverse & Composite Functions

A vector $\mathbf{v}$ is defined by its initial point $P = (-5, 3)$ and its terminal point $Q = (4, -2)$. What is the component form of vector $\mathbf{v}$?

a. $\langle -9, 5\rangle$

b. $\langle 9, -5\rangle$

c. $\langle -1, 1\rangle$

d. $\langle 1, -1\rangle$

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44 [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. Find the component form of $\mathbf{v}$.

a. $\langle 3, 4 \rangle$

b. $\langle 7, 10 \rangle$

c. $\langle -3, -4 \rangle$

d. $\langle 4, 3 \rangle$

45 [Challenging] • Target: Average rate of change • Cross-Lesson: 4.03 Parametric Equations & Modeling

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?

a. $\langle -5, 0 \rangle$

b. $\langle 5, 0 \rangle$

c. $\langle -3, 0 \rangle$

d. $\langle 3, 0 \rangle$

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